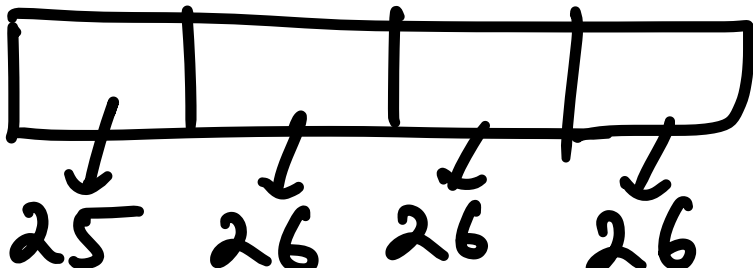


Permutation & Combination

① of all 4-letter words in the English alphabet, how many do not begin with an 'L'?

M-1



M-11

NOT begin with L

= Total - start with L

$$\text{Total} = (26)^4$$



choices

$$\Rightarrow 25 \times (26)^3$$

L			
---	--	--	--

 $= (26)^3$

$$\Rightarrow (26)^4 - (26)^3 \text{ Ans}$$

② A student ID is made up of 3 letters followed by (two digits). How many Student's ID's have some repetition?

Some Repititation of letters =

Total - NO Repetition of letters

$$(26)^3 \times (10)^2 = 26 \times 25 \times 24 \times 10 \times 9$$

③ Each user on a Computer System has a password, which is six to eight charac. long, where each char. is an upper case letter or a digit. Each



Password must contain at least one digit. How many possible passwords are there?

Passw ∴ $P_6 + P_7 + P_8$

$$(36)^6 - (26)^6 + (36)^7 - (26)^7 + (36)^8 - (26)^8$$

$P_6 \rightarrow$ at least one digit



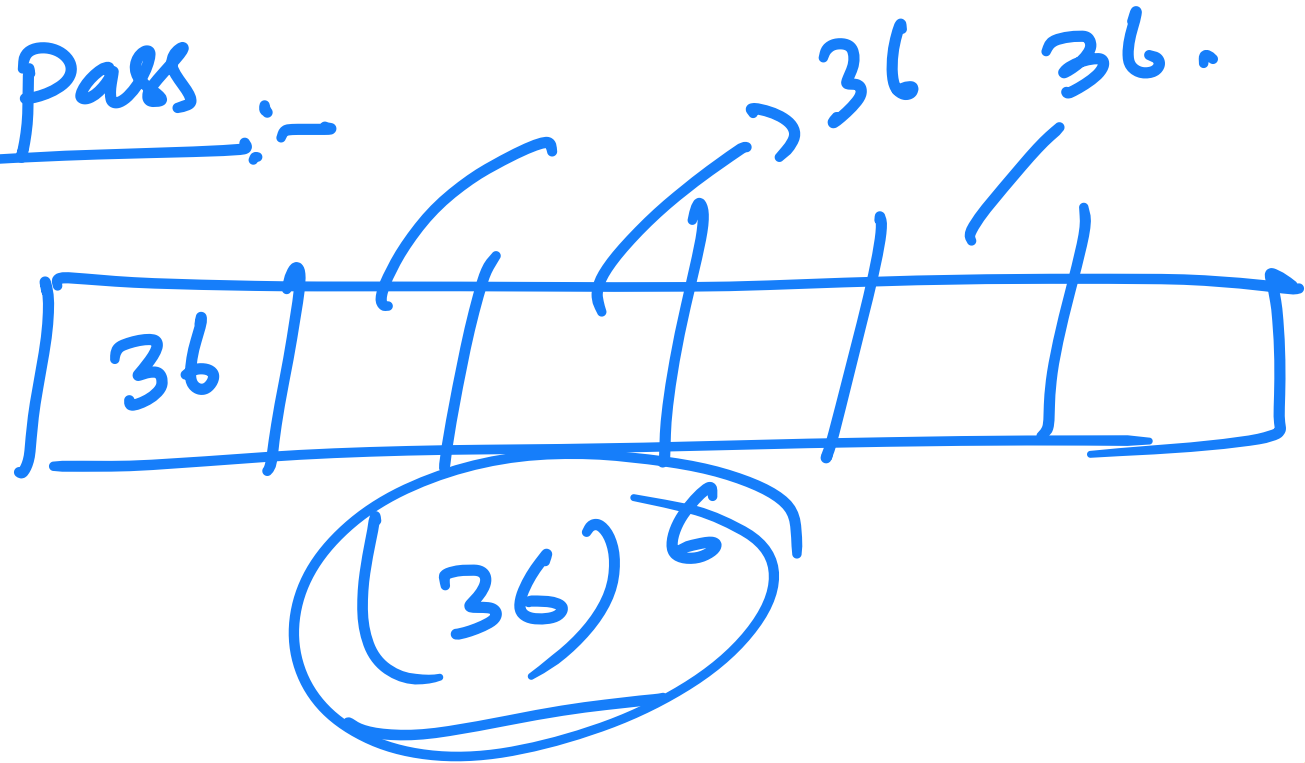
$$= \text{Total} - \text{No digit}$$

$$= (36)^6 - (26)^6$$

6-length pass:-

Total

$$+ (36)^6 =$$



..

Permutation & Combination

↓
Arrangement

a, b, c

a b c }
 a c b } two diff.
 Permutation

↓
Order

↓
Selection

Virat, Rohit, Pant

↓ select 2
Players

v, p }
 p, v } same
 selection

v, R
 R, p } order not
 matter



Does "Order matter"?

1. "My fruit salad is a Com. of apples, grapes and bananas" → order does not matter.
2. Set → order doesn't matter.
3. Team, Committee, Group →
4. Password → order matter



P&C

Order Arrangement

People, digits, numbers
Letters, Alphabet,
Colours etc.

selection

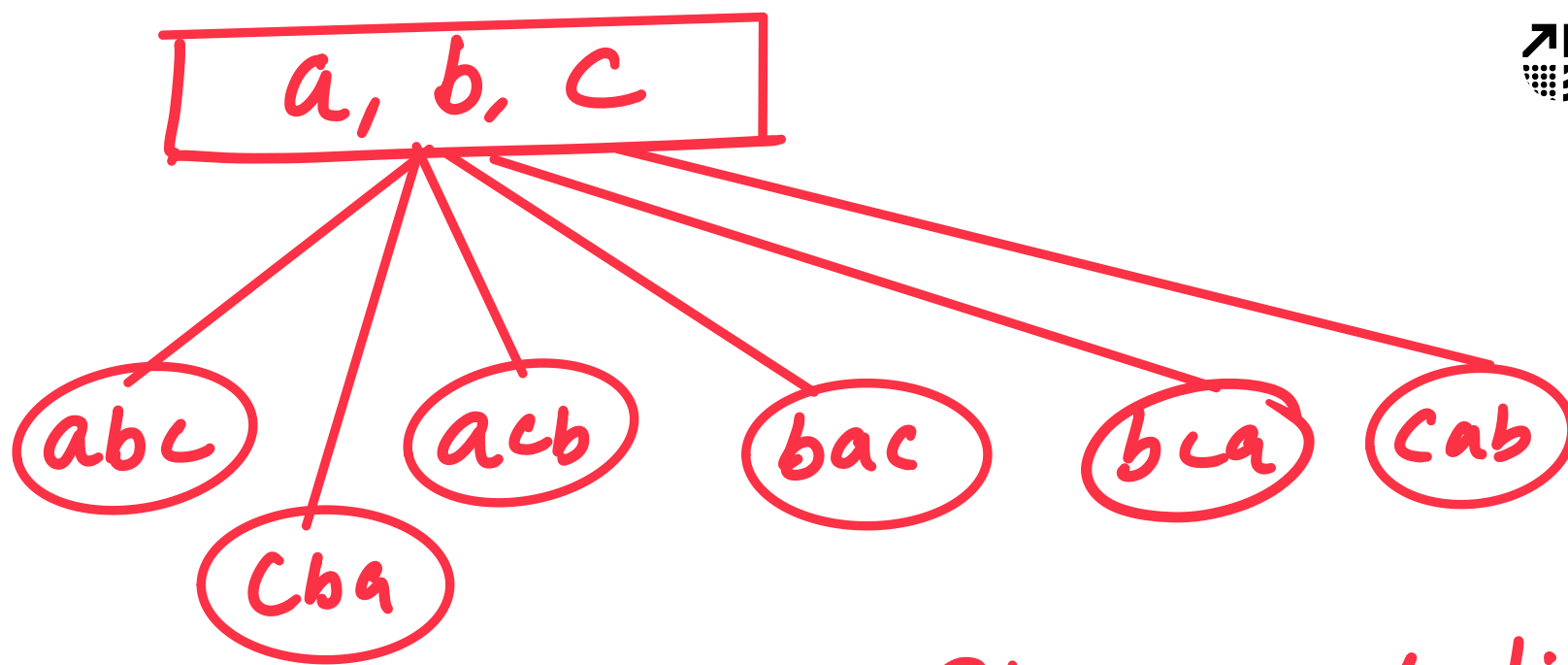
selection of menu,
food, clothes, Subj,
teams



P:- A permutation of a set of (distinct) objects is an ordering of the objects in row.

Q:- In how many diff. orders can three numbers finish a race if no ties are allowed?





There are six permutations.

Q. No. of Permutations/orderings/arrangements of 4 distinct elements are there?

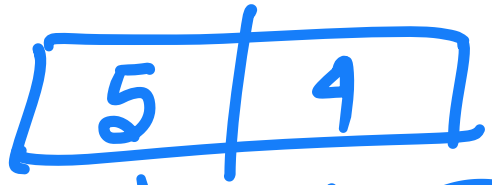




$$4 \times 3 \times 2 \times 1 = 4!$$

Q. No. of per/ord/arr of 2 distinct elements from a set of 5 elements are there?





$$5 \times 4 = 20$$

$${}^5P_2 = 20$$

Note

$${}_nP_r = \frac{n!}{(n-r)!}$$

$$= \frac{n(n-1)(n-2) \dots (n-r+1) \cancel{(n-r)!}}{\cancel{(n-r)!}}$$

$${}_nP_r = n \cdot (n-1)(n-2) \dots (n-r+1).$$

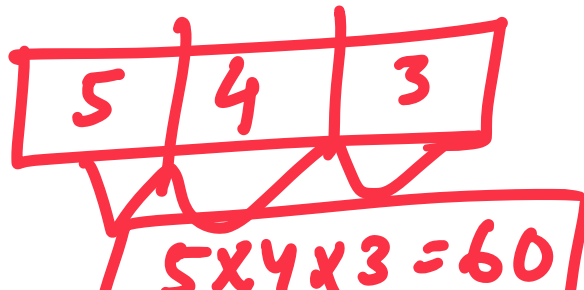


terms = 8

$${}^5P_2 = 20$$

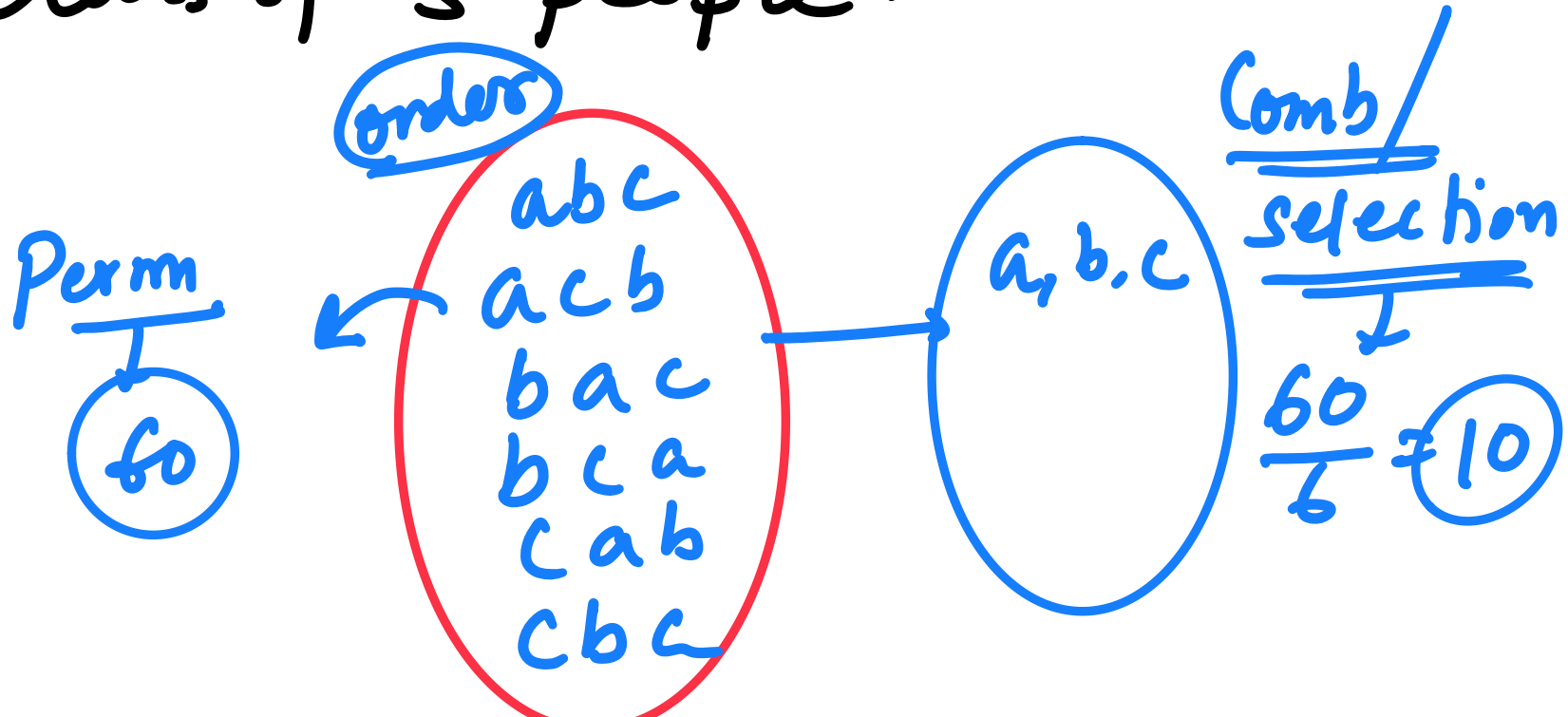
$${}^7P_4 = (7)(6)(5)(4).$$

Q. No. of ways to order 3 people from a class of 5 people.



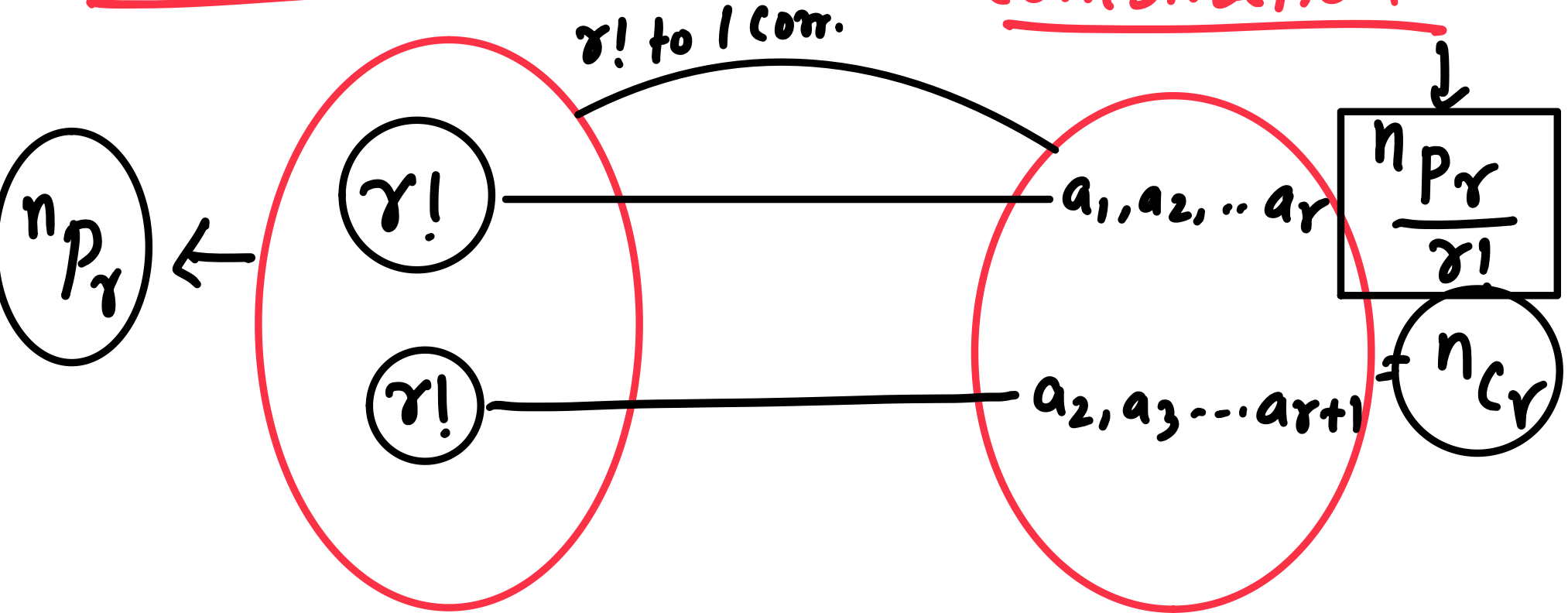
$$= {}^5P_3 = 5 \times 4 \times 3 = 60$$

Q. Num. of ways to select 3 people from a class of 5 people.



Permutation

Combination



Note : n -Distinct element

choose/select/com
↓
 r element

↓
ways
=
 nCr

Story behind this



$$\begin{aligned} n_{C_r} &= n_{\text{Choose}_r} \\ n_{P_r} &= n_{\text{Permute}_r} \end{aligned}$$

Note :-



$${}_nP_r = n \times (n-1) \times (n-2) \cdots (n-r+1)$$

r terms

$${}_nC_r = \frac{n(n-1) \cdot (n-2) \cdots (n-r+1)}{r!}$$

Application in Computer Science :-

1. Algorithm Design & Analysis.
2. Cryptography & security



3. Software Testing
4. AI/ML
5. Graph Theory/Network Analysis.

